

A STUDY ON THE FUNCTIONING OF THE HARITHA KARMA SENA IN THRISSUR DISTRICT, KERALA

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Contents

Chapter 1	4
INTRODUCTION.....	4
HARITHA KARMA SENA(HKS)	4
AIM OF HARITHA KARMA SENA.....	4
SUBJECT MATTER	5
OBJECTIVES	7
Chapter 2	8
METHODOLOGY	8
POPULATION AND SAMPLE.....	8
DIAGRAMS AND CHARTS	9
STATISTICAL TESTS	10
Chapter 3	12
STATISTICAL ANALYSIS	12
3.1 Descriptive Statistics	12
3.2 Statistical Tests.....	39
Chapter 4	45
SUMMARY	45
Chapter 5	53
REFERENCES.....	53

Chapter 1

INTRODUCTION

HARITHA KARMA SENA(HKS)

Haritha Karma Sena (HKS) is a professional team consisting of Green Technicians and Green Supervisors, mainly Kudumbashree Women, as it is a micro-entrepreneurial initiative under Kudumbashree mission the members of HKS are assigned with the responsibility of collection, transportation, processing, recycling/disposal, and management of waste materials in association with respective Local Self Governments (LSGs) and suchitwa mission. Haritha Karma Sena was launched in 2013 by the Government of Kerala as a part of the Haritha Keralam Mission, a comprehensive program to make Kerala a clean and green state. A minimum of 2 members forms the smallest unit of the Sena at the ward level of a local body. A micro-entrepreneurial group called the MEG is formed with 4-10 members. The members will collect non-biodegradable waste at doorsteps regularly. Besides, the MEGs are able to start businesses related to waste management with the help of Kudumbashree. The Mission will provide training and interest-free loans. The business can be of any type, such as upcycling or transforming the waste materials into new products, distribution and maintenance of bio-bins and biogas equipment, swap shops, farming using the by-products of bio-bins or even servicing the solar power panels. Educating households about waste management is also the duty of HKS. A Sena member has multiple streams of income such as monthly user fee, paid by each household, the price paid by the company that will acquire the plastic collected by HKS, and Businesses. They can also offer their services at marriages and public events by charging a fee.

As per the bylaw, the HKS is well organised. The MEG has a secretary and a president. Consortiums are formed at the local-body level consisting of all HKS members. In Corporations, the number of consortiums can be more than one. It has an executive committee that includes one member from each MEG. A consortium will have an elected secretary, president and treasurer as well. MEGs will pay a small amount every month to the consortium that will be used for the welfare of the HKS members. The consortium work as a grievance cell. It is to encourage and empower the HKS members to start their own businesses.

AIM OF HARITHA KARMA SENA

The mission aims to make Kerala a garbage-free state, promote sustainable waste management practices, create employment opportunities in the waste management sector and raise awareness among the public about the importance of waste management.

Non-organic waste is collected from households at regular intervals and delivered to the material collection facility known as MCF. From here, the waste is sorted and sent to the resource recovery facility or RRF. Then sorting waste enables combinations to be recycled. This is the modus operandi of Haritha karma Sena. By pulverizing the plastic waste in the resource recovery facility and using it for road tarring, it is possible to recycle the plastic waste and reduce the environmental

pollution through the work of the Haritha Karma Sena. In addition to this, Green Industries are trying to become self-sufficient by providing solutions for organic waste management to households and implementing other innovative business models such as green enterprises that make better products from waste. Haritha Keralam Mission, Suchitwa Mission, Clean Kerala Company and local bodies are all involved in this big project which is currently working in 676 panchayats of Kerala. In other local bodies, the work of Haritha karma Sena is in the preliminary stage. At present, more than 30,000 Kudumbashree women have been trained and is working in local institutions on an entrepreneurial model. Decentralized waste management and recycling through recycling of inorganic waste is growing as a solution to the country's waste problem and a better source of income for women.

This study was to understand the extent to which the Haritha keralam mission was able to implement and carry on with its missions, implementation of Haritha Karma sena and its applicability in managing the non-biodegradable waste from houses and establishments and also to know the public perspective about the HKS. For the study 11 panchayath's of Thrissur district was selected randomly. The students from the Dept. Of Statistics of St Thomas college (Autonomous) pursuing MSc was assigned for this investigation.

SUBJECT MATTER

1. Alagappanagar

Alagappanagar Grama Panchayat is divided into 17 wards. It consists of 10217 houses and 2245 establishments. The Haritha karma sena of this panchayat consists of 32 members. The amount of waste that they collect from both houses and establishments is 10 tons per month. 1 MCF and 4 mini MCFs are available. RRF is also working. The panchayat provides bio digester bins, and as of now, they are distributed in 430 houses or establishments.

2. Chelakkara

Chelakkara Grama Panchayat consists of 22 wards with a total of 11557 houses and 615 establishments. There are 45 members in Haritha Karma Sena. Totality of waste they have collected after its commencement is around 114 ton. Presently there are 2 MCFs and 1 is in progress also 9 mini MCFs are functioning in this panchayat.

3. Choondal

Total of 18 wards is present in the Choondal Panchayat. It consists of 9120 houses and 1250 establishments. The Haritha karma sena of this panchayat consists of 34 members. 1 MCF and 1 mini MCF are available. The panchayat provides bio digester bins.

4.Elavally

Elavally Grama Panchayat is divided into 16 wards. It consists of 9086 houses and 856 establishments. The Haritha karma sena of this panchayat consists of 31 members. The amount of waste that they collect from both houses and establishments is around tons per month. 1 MCF and 4 mini MCFs are available. 1 RRF is also working. The panchayat provides bio digesterbins.

5.Enghandiyur

Enghandiyur Grama Panchayat consists of 16 wards with a total of 10534 houses and 1791 establishments. Exactly 28 members are there in the Haritha Karma Sena. Totality of waste they collect is around 4 tons per month. 1 MCF and 4 mini MCFs are available. RRF is also working. The panchayat provides bio digesterbins, and as of now, they are distributed in 430 houses.

6.Kaiparambu

Kaiparambu grama panchayat office is located at Mundur in Thrissur district. Kaiparambu, Anjur, and Peramangalam villages are included in Kaiparambu panchayat. There are 18 wards in the panchayat and a total of 10589 houses and 2726 commercial places are present in the panchayat. There are 28 members in the Haritha karma sena. 1 MCF and 14 mini MCFs are functioning in the panchayat. The amount of waste collected is an average of 6 tons per month. The panchayat provides bio digester bins and around 1100 bins are already provided.

7.Madakkathara

Madakkathara Grama Panchayat is divided into 16 wards. It consists of 6919 houses and establishments. The Haritha karma sena of this panchayat consists of 32 members. The amount of waste that they collect from both houses and establishments is around tons per month. 4 MCFs are functioning. The panchayat provides biodigester bins.

8.Mulagunnathukavu

Mulagunnathukavu Grama Panchayat is divided into 14 wards. It consists of 6547 houses and 632 establishments. The Haritha karma sena of this panchayat consists of 20 members. The amount of waste that they collect from both houses and establishments is around 32 tons per month. 1 MCF and 17 mini MCFs are available. The panchayat provides bio digesterbins.

9.Nadathara

Nadathara Grama Panchayat is divided into 17 wards. It consists of 9521 houses and 607 establishments. The Haritha karma sena of this panchayat consists of 32 members. The amount of waste that they collect from both houses and establishments is close to 8 tons per month. 1MCF and 3 mini MCFs are available. The panchayat provides biodigesterbins.

10.Nenmanikkara

There are 15 wards in the panchayat. The amount of waste that they collect from both houses and establishments is 6 ton per month. 1 MCF and 2 mini MCF are available. Mrs. shaiji got the award for the highest paid Haritha karma sena member who is working on Nenmanikkara panchayat. Also Nenmanikkara panchayat have got an Award for the most segregated Panchayat and Harita Karma Sena Award for Model Panchayat for close and systematic work .

11.Vadekkekad

Vadakkekad Grama Panchayat is divided into 16 wards. It consists of houses and establishments. The Haritha karma sena of this panchayat consists of 35 members. The amount of waste that they collect from both houses and establishments is 5 tons per month. 1 MCF and 17 mini MCFs are available and functioning. The panchayat provides biodigester bins, and as of now, they are distributed in 399 houses.

OBJECTIVES

- To understand the extent of activities of Harita Karma Sena in every panchayat
- To study how satisfied the Harita Karma Sena members are with the way they are treated by the public
- To learn about the physical challenges faced by Harita Karma Sena members after taking up this job
- To understand how satisfied are the public with the activities of Harita Karma Sena
- Knowledge possessed by the public about Haritha Kerala Mission and execution of waste management after its commencement.
- To understand the extent to which the distribution of biodigester bin and its efficient use is being done in each panchayat

Chapter 2

METHODOLOGY

POPULATION AND SAMPLE

The collection of all units of a specified type in a given region at a particular point or a period of time is termed as a population. One or more sampling units are selected from the population according to some specified procedure. A sample consists of only a portion of the population units. Such a collection of units is called the sample. The total number of sampling units in the population is the population size, generally denoted by N . The population size can be finite or infinite (N is large)

Population covered

The group of individuals (elements) defined according to the aim of the survey is called population or universe. The word population is used to denote the aggregate from which the sample is chosen. The population should be defined in clear and unambiguous terms. In this survey the population selected is Grama Panchayats in Thrissur district. The data is collected using questionnaires.

Sampling procedure

The selection of units in the sample from a population is governed by the laws of chance or probability. The probability of selection of units are random.

Determination of sample size

The data was collected from 11 panchayath's of Thrissur district which are selected at random. Three questionnaires were prepared to collect the information from the public about HKS service, from the Haritha karma sena members and then from the panchayat officials regarding the functioning of Haritha karma sena. The information about the Haritha karma Sena service was collected from 75-80 citizens of each panchayat. Details regarding debris collections and Haritha karma sena are collected from a panchayath official. The overall perspective of Haritha karma sena members regarding their employment was also inquired.

Source of statistical data

In the present study the information was collected using a questionnaire. A set of questions is prepared according to the objectives and it is distributed among the sample of the population. The respondent fills the given questionnaire. Hence questionnaire method is used for collection of data. Survey data obtained from the questionnaire is classified, tabulated and analyzed. Simple analytical tool like tables, bar diagrams, pie chart, cross tabulation are used for the classification and presentation of the data finally. We performed statistical tests using the R software.

Preparation of questionnaire

A questionnaire is a research instrument that consists of a number of questions that the respondents have to answer in a set format. A distinction is made between open-ended and close-ended questions. An open-ended question asks the respondent to formulate his own answers, where a close-ended question has the respondent to pick up the answer from a given number of options. The response option for a close-ended question should be exhaustive and mutually exclusive.

DIAGRAMS AND CHARTS

A diagram is a visual form of presentation of statistical data, highlighting basic facts and relationship. Though diagrams do not add any new meaning to the statistical facts, yet they exhibit the result more clearly.

Bar Diagram

A bar graph is a graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. It is a method of presenting data in which frequency are displayed along one axis and categories of variable along the frequency being represented by the bar length. The bars can be plotted vertically or horizontally.

Pie Diagram

A special chart that uses “pie slices” to show relative sizes of data. It is a really a good way to show relative sizes. A pie chart is a specialized chart used in statistics.

Cross tabulation

Arranged cross tabulation is a combination of 2 or more frequency tables such that each cell in the resulting table represents a unique combination of specific value of cross tabulates variables. Thus, cross tabulations allow us examine frequencies of observations that belong to specific categories on more than one variable. By examining these frequencies, we can identify relations between cross tabulation variables. Only categorical (normal) variable or variables with a relatively small

number of different meaningful values should be cross tabulated. The simplest form of cross tabulation is the 2×2 table where 2 variables are “crossed”, and each variable has only two distinct values.

LIKERT -TYPE SCALES

Likert questions can help you ascertain how strongly your respondents agree to a particular statement. Such types of questions also help you assess how your customers feel towards a certain issue, product or service. For example, Unilever’s products have to improve on quality - (1) Strongly Agree (2) Agree (3) Neither Agree or Disagree (4) Disagree (5) Strongly Disagree.

STATISTICAL TESTS

Chi-square test

The chi- squared test is a test of association. It is used when we have nominal data in form of frequency, and when each and every observation is independent of all the other observations. Studies often collect data on categorical variables that can be summarized as a series of counts. These counts are commonly arranged in a tabular format known as a contingency table. The chi square test statistic can be used to evaluate whether there is an association between m rows and n columns in a contingency table. The equation used to conduct chi – square test of independence in statistical analysis.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(m-1, n-1)}$$

Where, O_i = Observed frequency

E_i = Expected frequency

m, n - number of rows and columns of the contingency table.

KOLMOGOROV- SMIRNOV TEST

The Kolmogorov–Smirnov one-sample test is a procedure to examine the agreement between two sets of values. This test is used for the Normality checking. Whether the population is normally distributed. The Kolmogorov– Smirnov one-sample test compares two cumulative frequency distributions. A cumulative frequency distribution is useful for finding the number of observations above or below a particular value in a data sample. It is calculated by taking a given frequency and adding all the preceding frequencies in the list. In other words, it is like making a running total of

the frequencies in a distribution. The null hypothesis states that the observed sample has an approximately normal distribution. The research hypothesis states that the observed sample does not approximately resemble a normal distribution.

Test statistic

$$Z = \sqrt{n} \max(|D|, |\tilde{D}|)$$

Where 'n' is the sample size and D and $\tilde{D}$ divergence between values from cumulative frequency distributions.

THE FISHER'S EXACT TEST

Fisher's exact test is practically applied only in analysis of small samples but actually it is valid for all sample sizes. While the chi-squared test relies on an approximation, Fisher's exact test is one of exact tests. Especially when more than 20% of cells have expected frequencies < 5 , we need to use Fisher's exact test because applying approximation method is inadequate. Fisher's exact test assesses the null hypothesis of independence applying hypergeometric distribution of the numbers in the cells of the table. Many packages provide the results of Fisher's exact test for 2 x 2 contingency tables but not for bigger contingency tables with more rows or columns. For example, the SPSS statistical package automatically provides an analytical result of Fisher's exact test as well as chi-squared test only for 2 x 2 contingency tables.

MANN- WHITNEY U TEST

Mann-Whitney U test is a nonparametric test of null hypothesis that, that, for randomly selected values X and Y from two populations, the probability of X being greater than Y is equal to the probability of Y being greater than X. Let $X_1, \dots, X_{n_1}$ be an i.i.d. sample from X, and $Y_1, \dots, Y_{n_2}$ an i.i.d. sample from Y, and both samples independent of each other. The corresponding Mann–Whitney U statistic is defined as the smaller of:

$$U_1 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

$$U_2 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

With R_1, R_2 being the sum of the ranks in groups 1 and 2, after pooling all samples in one set and where the smallest value obtains rank 1 and so on.

Chapter 3

STATISTICAL ANALYSIS

In order to interpret the data in a scientific manner according to our objectives, the data is subjected to statistical analysis. Descriptive statistics try to describe the relationship between variables in a sample or population and provides a summary of the data in the form of diagrams and charts. In inferential statistics the data are analyzed by performing hypothesis tests like chi squared test, proportional test etc.

3.1 Descriptive Statistics

3.1.1. To Study How Satisfied The Haritha Karma Sena Members Are With The Way They Are Treated By The Public

a. Satisfaction to the public behavior

Here we are looking into the satisfaction of Haritha karma sena members towards the public reaction

Table 3.1.1.a Satisfaction to the public behavior

Haritha Karma Sena satisfaction	Frequency	Percent
Very satisfied	25	13.095
Satisfied	63	36.842
Neutral	28	16.374
Dissatisfied	50	29.240
Very Dissatisfied	5	2.924
Total	171	100.000

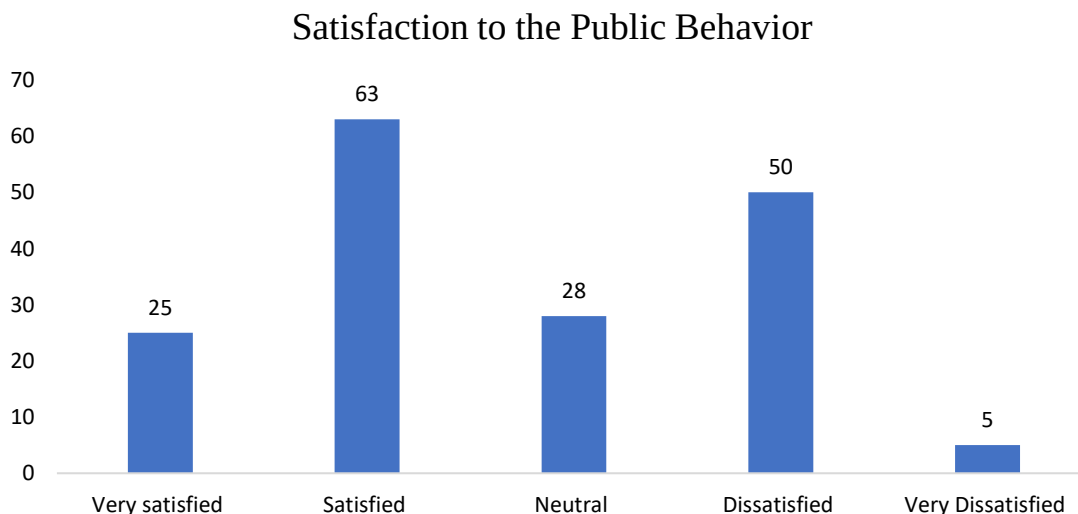


Figure 3.1.1.a Satisfaction to the public behavior

Inference

From the graph we can see that 12% of the respondents are very satisfied and 36% are satisfied towards the public reaction.16% have a neutral opinion about public reaction and 29% are dissatisfied.

b. Public Reluctance

Here we are checking the reluctance of public to the Haritha karma sena activities.

Table 3.1.1.b Public reluctance

Reluctance from public	Frequency	Percent
No	44	25.731
Yes	127	74.269
Total	171	100.000

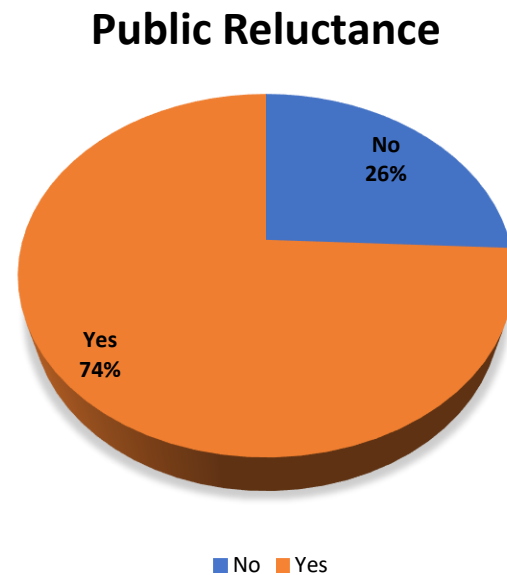


Figure 3.1.1.b Public reluctance

Inference

From the above graph we can see that 74% of the respondents have experienced reluctance from public while collecting waste.

c. Bad Experience from Public

Here we are looking whether the Haritha karma sena members had any bad experience from the public.

Table 3.1.1.c Bad experience from public

Any bad experience from public	Frequency	Percent
No	41	23.977
Yes	130	76.023
Total	171	100.000

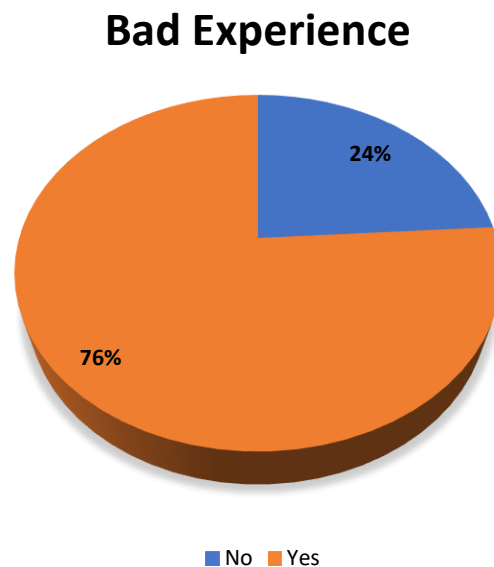


Figure 3.1.1.c Bad experience from public

Frequency table of, from where they had a bad experience is given below

Table 3.1.1.c Where they had a bad experience

	Frequency	Percent
From Houses	108	82.443
From Organizations	22	17.557
Total	130	100.000

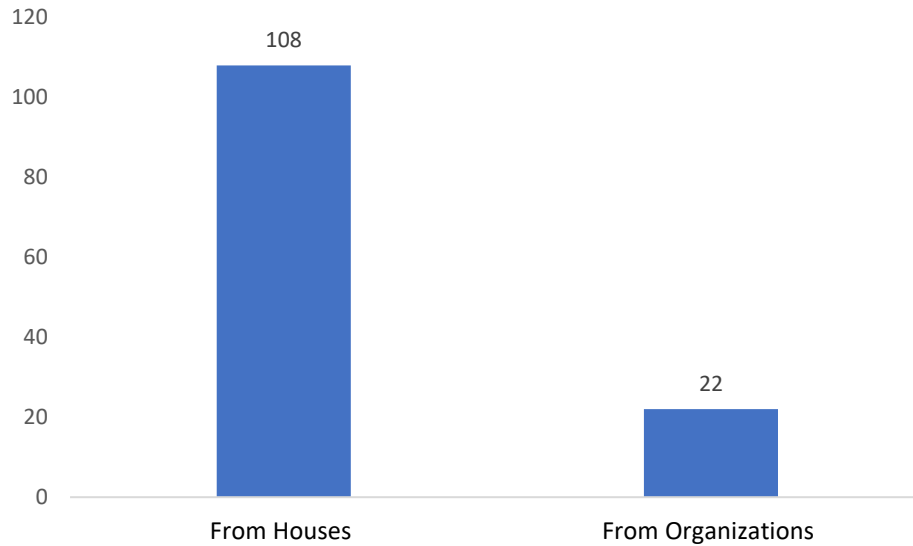


Figure 3.1.1.c Where they had a bad experience

Inference

From the above graph we can see that 76% of the respondents had bad experience from the public and it is mostly from houses.

3.1.2. To Learn About the Physical Challenges Faced by Haritha Karma Sena Members After Taking Up This Job

a. Age distribution of Haritha karma sena

Here we are looking the age of Haritha karma sena members.

Table 3.1.2 .a Age distribution of Haritha karma sena

Age	Frequency	Percent
Till 40	26	15.205
40-50	63	36.842
50-60	66	38.012
60-70	16	9.357
Total	171	100.000

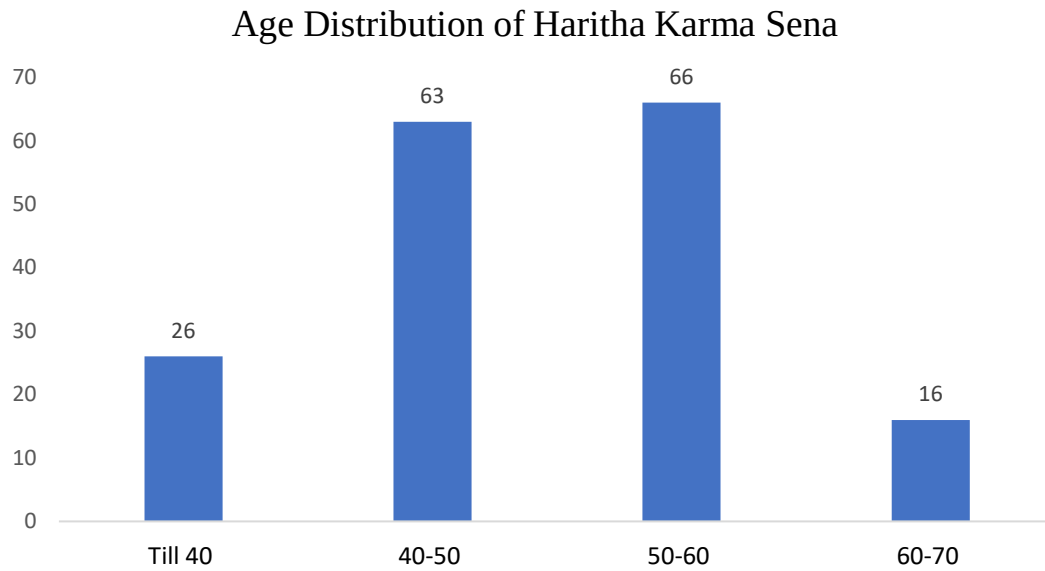


Figure 3.1.2.a Age distribution of Haritha karma sena

Inference

From the graph we can see that most of the workers belongs to the age category 50 - 60

b. Previous working experience

Here we are checking the previous working experience of Harita karma sena members.

Table 3.1.2.b . Previous working experience

have you done any job before	Frequency	Percent
No	23	13.450
Yes	148	86.550
Total	171	100.000

Haritha karma sena members who have done other jobs before

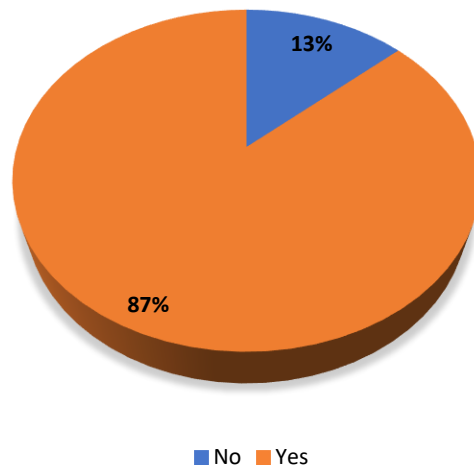


Figure 3.1.2.b . Previous working experience

Inference

From the above graph we can see that 87% of the respondents have been engaged in other jobs before becoming a haritha karma sena member.

c. Hospital Visiting Frequency

Here we are checking how many times a Haritha karma sena member is visiting the hospital in a month.

Table 3.1.2.c Hospital Visiting Frequency

frequency of visiting doctor	Frequency	Percent
Not visiting	69	40.351
Once	60	35.088
Twice	28	16.374
Three times	10	5.848
Four times	2	1.170
Five times	2	1.170
Total	171	100.000

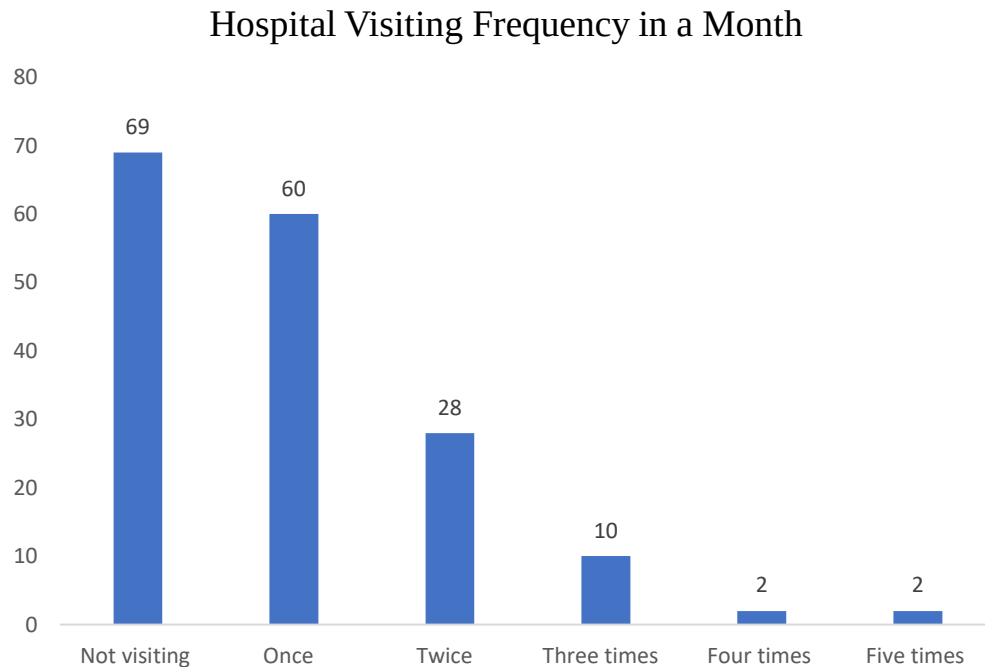


Figure 3.1.2.c Hospital Visiting Frequency

Inference

40.4% members do not visit hospital in a month, 35% members visit hospital once in a month and only around 1% members visit 4 or 5 times in a month.

d. Increase in hospital visiting frequency

Here we are looking if there is any increase in the frequency of visiting hospital after taking up this job.

Table 3.1.2.d Increase in hospital visiting frequency

Increase in hospital visiting frequency	Frequency	Percent
No	125	73.099
Yes	46	26.901
Total	171	100.000

Increase in Hospital visiting

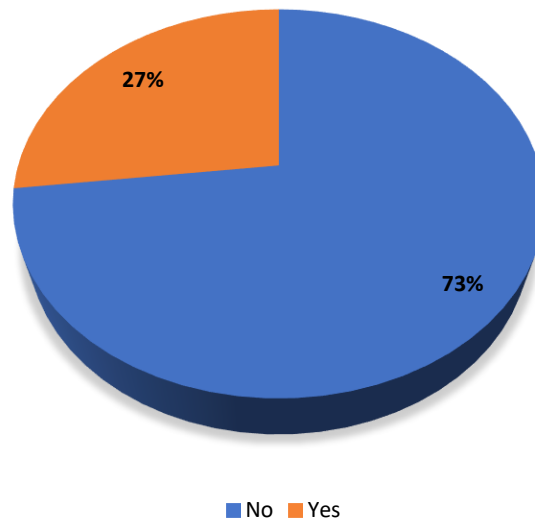


Figure 3.1.2.d Increase in hospital visiting frequency

Inference

The above graph shows that for 27% of the respondents there is an increase in hospital visiting frequency after engaging in this work. 73% of the respondents does not feel an increase in hospital visiting frequency.

e. Health Issues After taking up this job

Here we are checking whether there is any health issues that are caused by this job

Table 3.1.2.e Health Issues After taking up this job

Health issues after engaging in this work	Frequency	Percent
No	69	41.520
Yes	102	58.480
Total	171	100.000

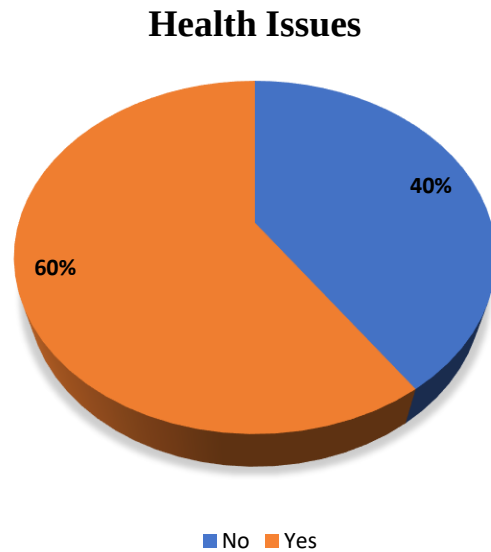


Figure 3.1.2.e Health Issues After taking up this job

Inference

From the graph we can see that 60% of the respondents experienced health issues after engaging in this work.

f. The Health Issue specified by the HKS members

If there is any health problems, here we are checking what are those.

Table 3.1.2.f Health Issue specified by the HKS members

can you specify the health issue	Frequency	Percent	Valid Percent
Back Pain	16	9.357	15.686
Cough	14	8.187	13.725
Others	37	21.637	36.275
Skin Allergy	35	20.468	34.314

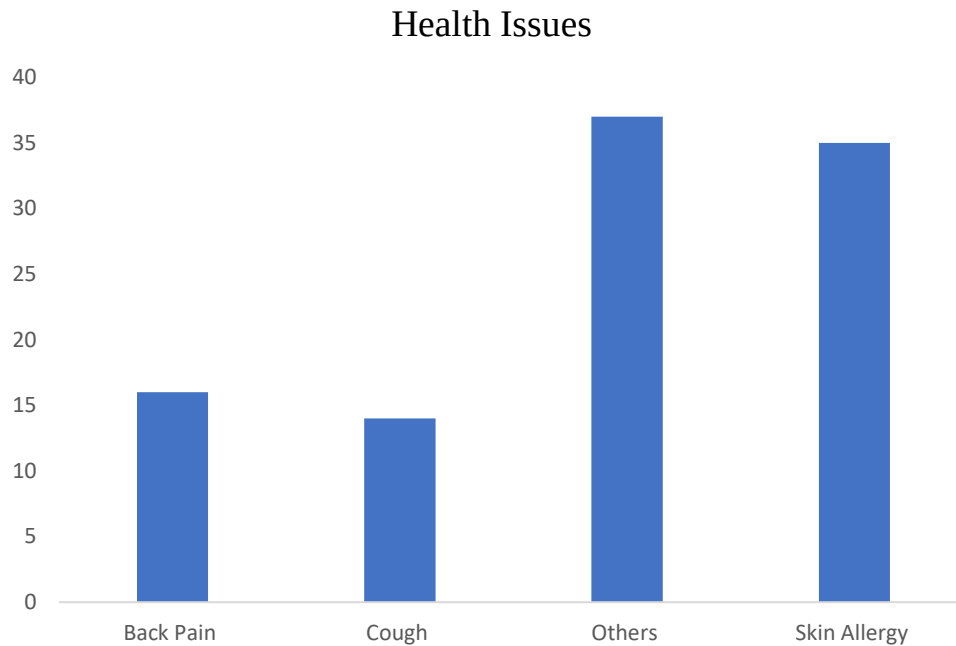


Figure 3.1.2.f Health Issue specified by the HKS members

Inference

Out of the people who have a health issue 34% have skin allergy problems, 15% have back pain, 13.7% have cough and 36% have other health issues.

g. Number of Working days in a month

Here we are looking the working days in a month of Haritha karma sena members.

Table 3.1.2.g Number of Working days in a month

working days	Frequency	Percent
Below 20	49	28.655
20-25	26	15.205
25-30	96	56.140
Total	171	100.000

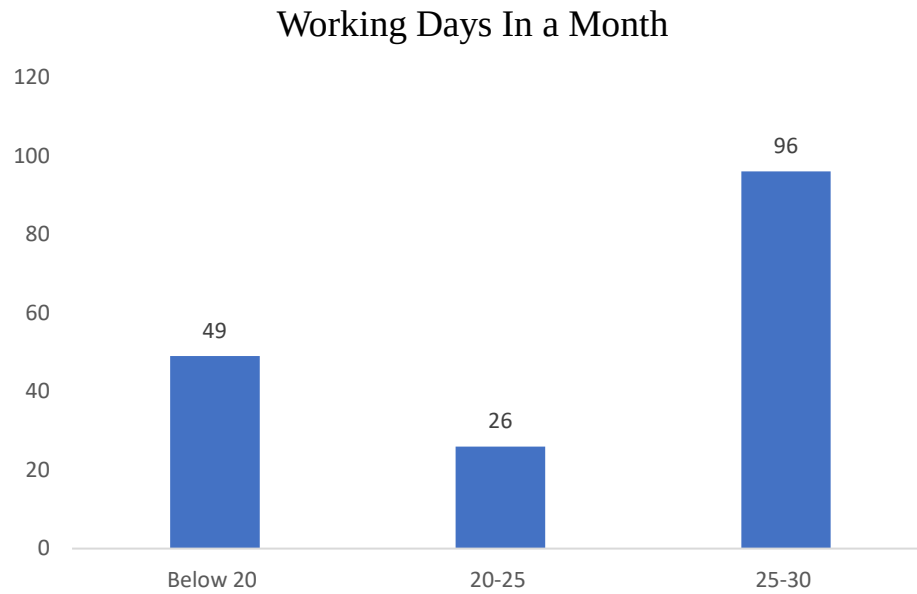


Figure 3.1.2.g Number of Working days in a month

Inference

From the above graph we can see that most of the respondents work 25-30 days in a month.

h. Working hours in a day

Here we are looking how many hours in a day Haritha karma sena work.

Table 3.1.2.h Working hours in a day

Working hours	Frequency	Percent
6 hours	2	1.170
7 hours	24	14.035
8 hours	114	66.667
9 hours	31	18.129
Total	171	100.000

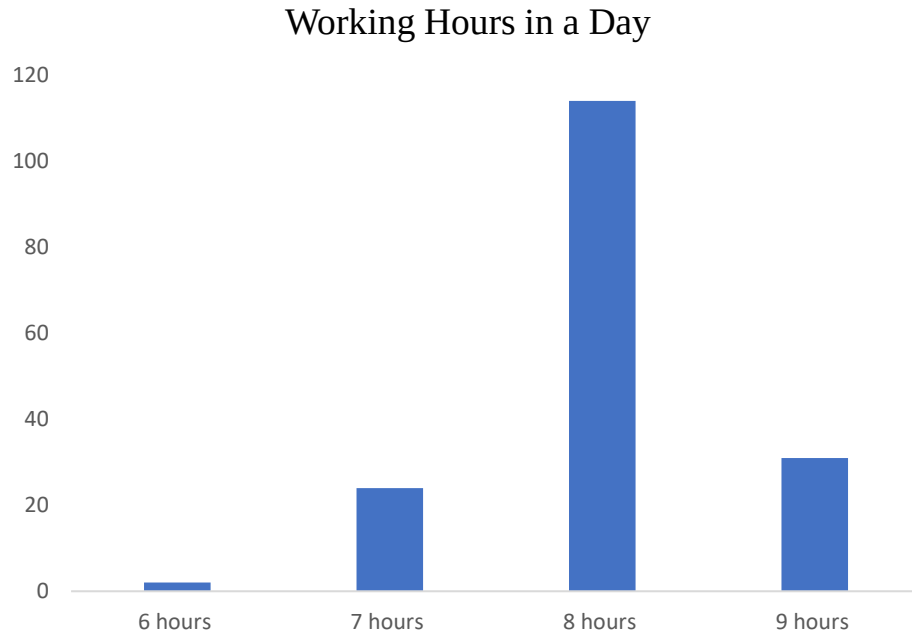


Figure 3.1.2.h Working hours in a day

Inference

From the above graph we can see that most of the respondents work 8 hours per day.

3.1.3. To Understand How Satisfied Are The Public With The Activities Of Haritha Karma Sena

a. Monthly arrival of Haritha karma sena

Here we are checking if Haritha karma sena members are visiting houses and organizations every month.

Table 3.1.3.a Monthly arrival of Haritha karma sena

Do you receive service of HKS each month?	Frequency	Percent
No	23	4.093
Yes	539	95.907
Total	562	100.000

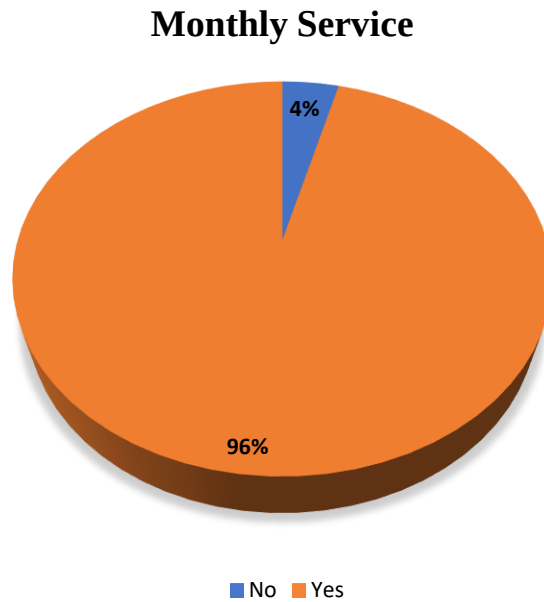


Figure 3.1.3.a Monthly arrival of Haritha karma sena

Inference

96% of public are saying that Haritha karma sena visits every month and remaining 4% are saying that they are not visiting.

b. Satisfaction in Haritha Karma sena services

Here we are checking whether the public is satisfied in the service of Haritha karma sena.

Table 3.1.3.b Satisfaction in Haritha Karma sena services

Are you satisfied in the service of HKS?	Frequency	Percent
Very satisfied	194	34.520
Satisfied	252	44.840
Neutral	84	14.947
Dissatisfied	28	4.982
Very Dissatisfied	4	0.712
Total	562	100.000

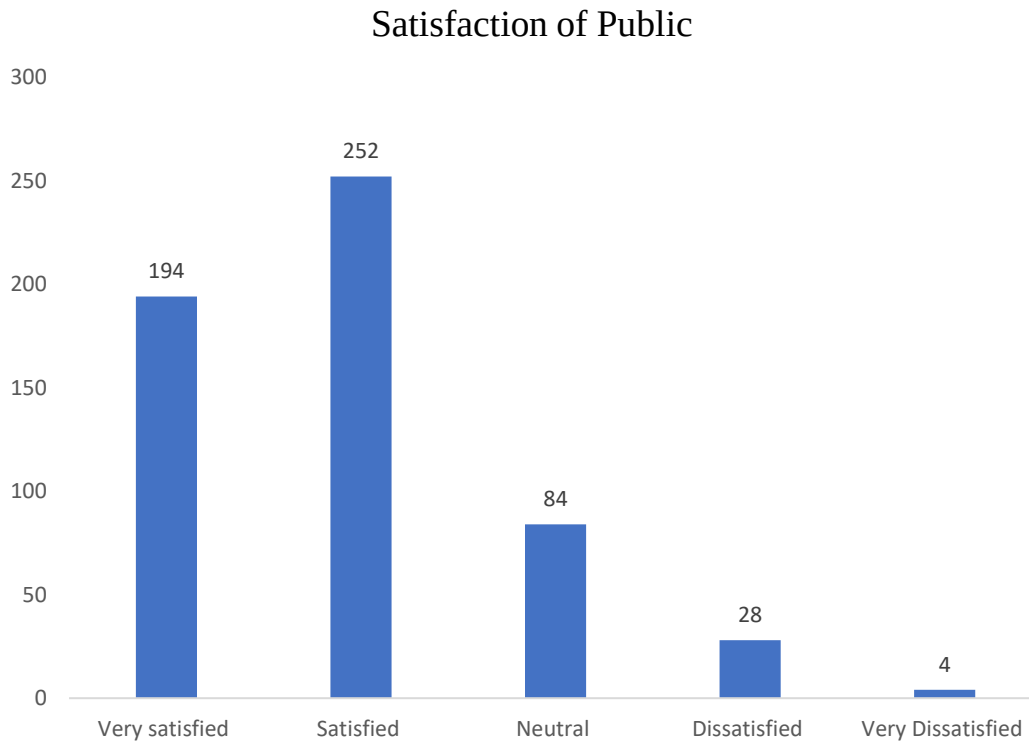


Figure 3.1.3.b Satisfaction in Haritha Karma sena services

Inference

Around 78% public are satisfied with the service of Haritha karma sena, around 6% are dissatisfied and others are neutral.

c. Bad Experience from Haritha karma sena

Here we are checking if the public had any bad experience from Haritha karma sena.

Table 3.1.3.c Bad Experience from Haritha karma sena

Did you have any bad experience from HKS?	Frequency	Percent
No	522	92.883
Yes	40	7.117
Total	562	100.000

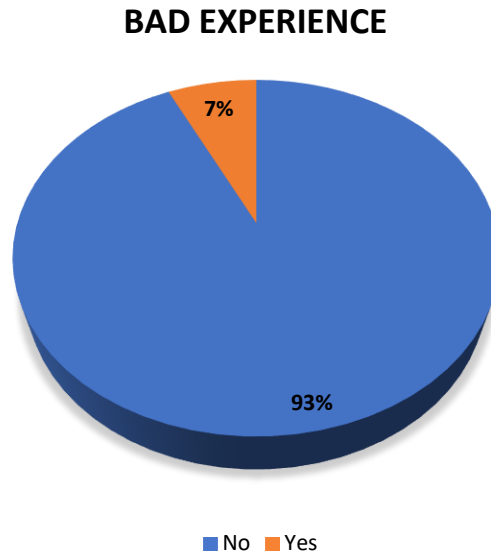


Figure 3.1.3.c Bad Experience from Haritha karma sena

Inference

Only 7% people had a bad experience from Haritha karma sena, 93% did not have any bad experience.

d. Illegal Dumping of waste

Here we are checking if the public see any illegal dumping of wastes still on the road sides.

Table 3.1.3.d Illegal Dumping of waste

Illegal dumping of garbage even after starting HKS			Frequency	Percent
No			427	75.979
Yes			135	24.021
Total			562	100.000

Illegal Dumping of waste

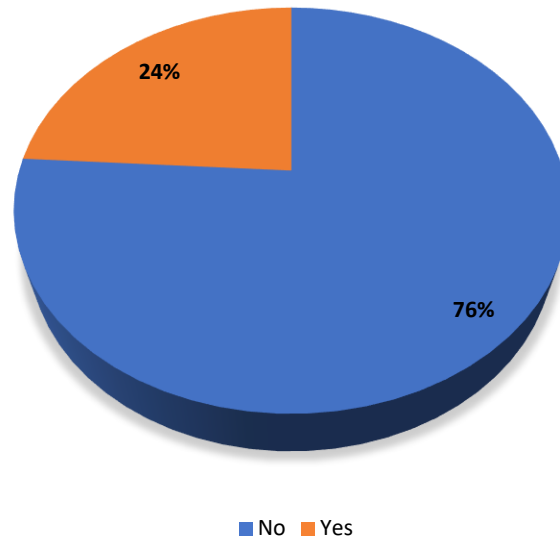


Figure 3.1.3.d Illegal Dumping of waste

Inference

24% says there is still waste dumping on road sides and 76% says no.

e. Misbehavior of people to Haritha karma sena

Here we are looking if anyone have seen others treating Haritha karma sena in a bad way.

Table 3.1.3.e Misbehavior of people to Haritha karma sena

Did you notice anyone behaving badly to HKS?	Frequency	Percent
No	516	91.815
Yes	38	6.762
No Response	8	1.423
Total	562	100.000

Inference

93% are saying they did not notice anyone misbehaving to Haritha karma sena and 7% are saying yes.

f. Knowledge about the Monthly Arrival of Haritha Karma Sena

Here we are looking whether the public is aware about the date of monthly arrival of Haritha Karma sena.

Table 3.1.3.f Knowledge about the Monthly Arrival of Haritha Karma Sena

Information About monthly arrival of HKS members	Frequency	Percent	Valid Percent	Cumulative Percent
No	259	46.085	46.085	46.085
Yes	303	53.915	53.915	100.000
Total	562	100.000		

For those who responded yes, here we are checking from whom they are informed about the arrival of Haritha karma sena members.

Table 3.1.3.f If yes, from whom they are informed

If yes, from whom you are informed	Frequency	Percent
Asha Worker	19	6.271
HKS	76	25.083
Kudumbasree	10	3.300
Others	23	7.591
Panchayat Member	175	57.756
Total	303	100.000

Monthly Arrival Information of HKS

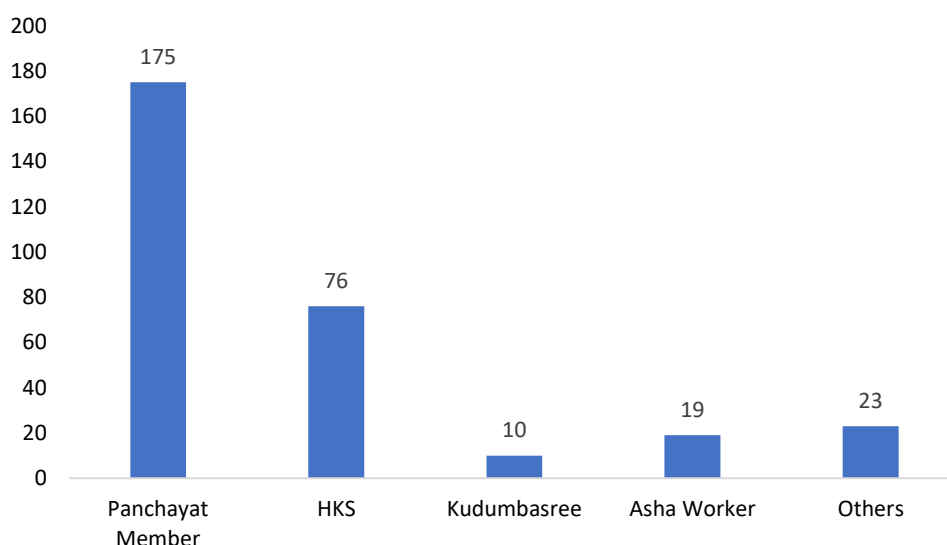


Figure 3.1.3.f If yes, from whom they are informed

Inference

54% of the public are early informed about the monthly arrival of Haritha karma sena and 58% of them are informed by panchayat member, 25% are informed by HKS itself, 6% are by asha workers, 3% are by kudumbasree and 8% are informed by other people.

3.1.4. Knowledge Possessed By The Public About Haritha Kerala Mission And Execution Of Waste Management After Its Commencement.

a. Fines on illegal dumping of waste

Here we are checking if the public is aware about the fines on illegal waste disposal.

Table 3.1.4.a Fines on illegal dumping of waste

Do you know at least 3 fines related waste disposal?	Frequency	Percent
No	196	34.875
Yes	366	65.125
Total	562	100.000

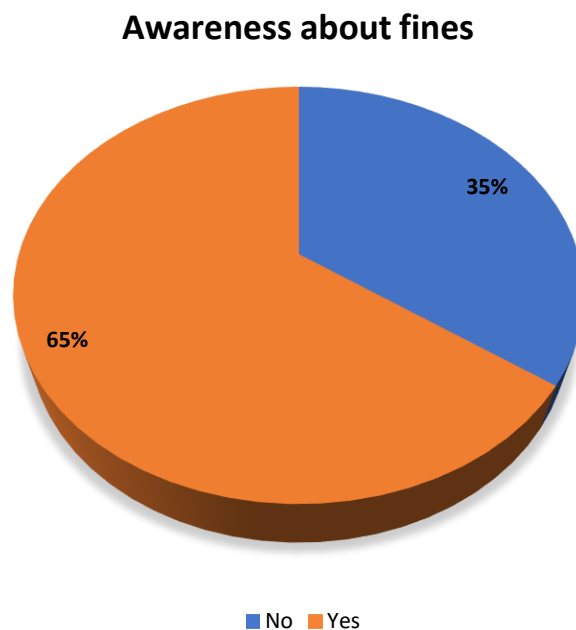


Figure 3.1.4.a Fines on illegal dumping of waste

Inference

65% of public are aware about the fines and 35% are not aware.

b. Awareness about goals of Haritha keralam mission

Here we are checking if the public is aware about the goals of Haritha keralam mission.

Table 3.1.4.b Awareness about goals of Haritha keralam mission

Are you aware about the goals of Haritha Kerala Mission?	Frequency	Percent
No	205	36.477
Yes	357	63.523
Total	562	100.000

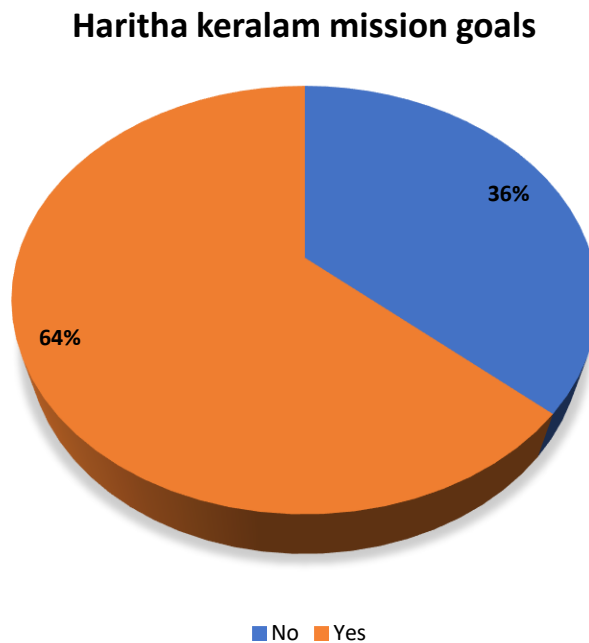


Figure 3.1.4.b Awareness about goals of Haritha keralam mission

Inference

64% of public are aware and remaining 36% are not aware about the goals of Hartha Keralam Mission.

c. Awareness given by Haritha karma sena

Here we are checking if the Haritha karma sena members are giving awareness to the public.

Table 3.1.4.c Awareness given by Haritha karma sena

Do you try to make people aware about your services ?	Frequency	Percent
No	5	2.924
Yes	166	97.076
Total	171	100.000

Awareness given by Haritha karma sena

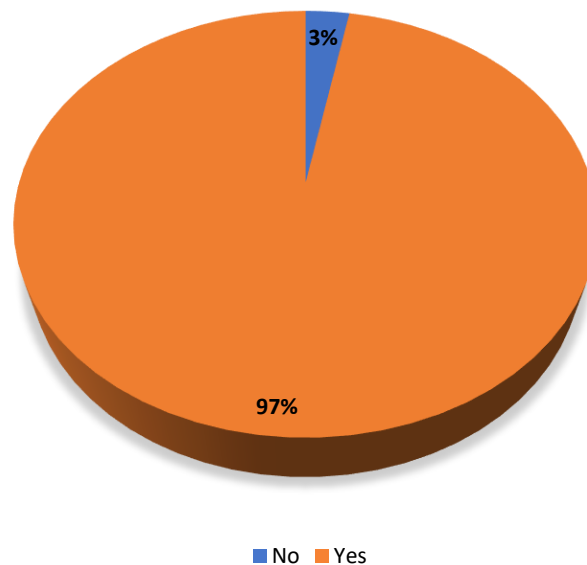


Figure 3.1.4.c Awareness given by Haritha karma sena

Inference

97% of Haritha Karma sena members are making the public aware of their services.

3.1.5. To Analyse the Haritha Mithram App Usage of Haritha karma sena members

a. Efficient use of Haritha mithram app

Here we are checking whether the HKS members are using the app effectively.

Table 3.1.5 Efficient use of Haritha mithram app

Efficient use of Haritha mithram App	Frequency	Percent
No	53	30.994
Yes	118	69.006
Total	171	100.000

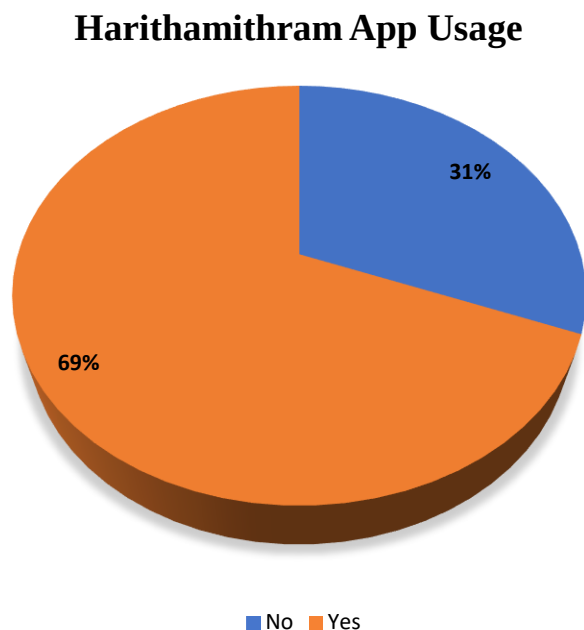


Figure 3.1.5.a Efficient use of Haritha mithram app

Inference

69% of HKS are using the app efficiently and 31% are not.

b. Technical Issues while using app

Here we are checking if there is any technical issues take place while using the app.

Table 3.1.5.b Technical Issues while using app

Technical Errors While Using the App	Frequency	Percent
No	26	15.205
Yes	138	80.702
No response	7	4.094
Total	171	100.000

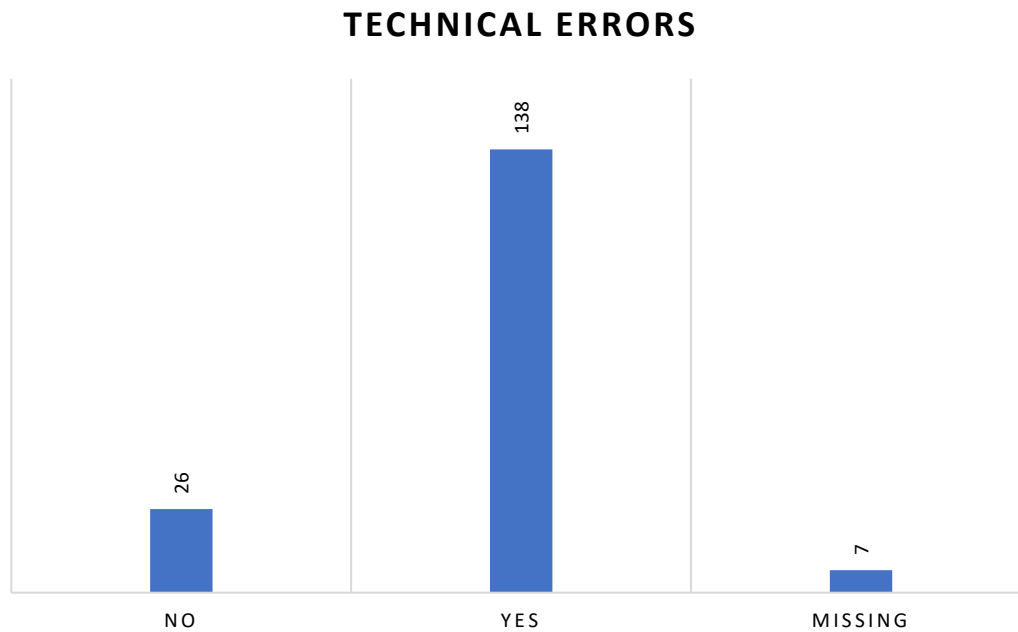


Figure 3.1.5.b Technical Issues while using app

Inference

81% are saying there is technical issues and 15% are saying that there are no issues, and the issues are specifically related to network connectivity.

c. Ease of work after using Haritha mithram app

Here we are checking if the app is making their job easy or not.

Table 3.1.5.c Ease of work after using Haritha mithram app

Does the app make your job easy	Frequency	Percent
No	36	21.053
Yes	107	62.573
No response	28	16.374
Total	171	100.000

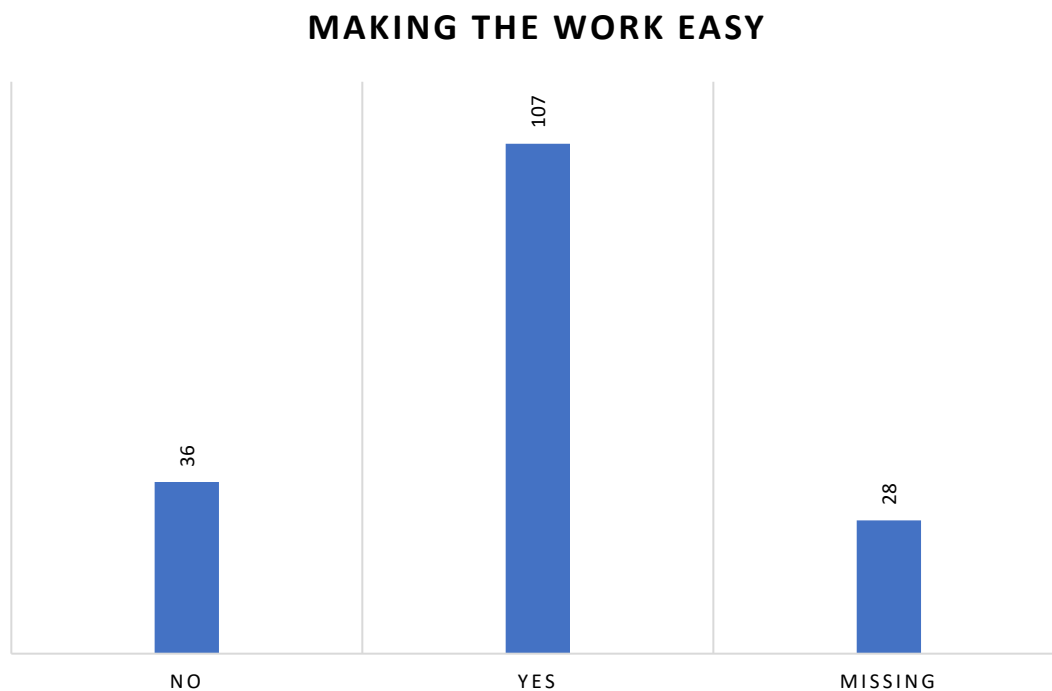


Figure 3.1.5.c Ease of work after using Haritha mithram app

Inference

63% are saying app makes the work easy while 21% are saying no.

3.1.6. To Understand The Extent To Which The Distribution Of Biodigester Bin And It's Efficient Use Is Being Done In Each Panchayat

a. Bio digester bin distribution in each panchayath

Here we are looking how many panchayat has given the biodigester bin to people.

Table 3.1.6.a Bio digester bin distribution in each panchayath

did you give the bio bin /port to people for the diposal of bio digestable wastes	Frequency	Percent
No	1	9.091
Yes	10	90.909
Total	11	100.000

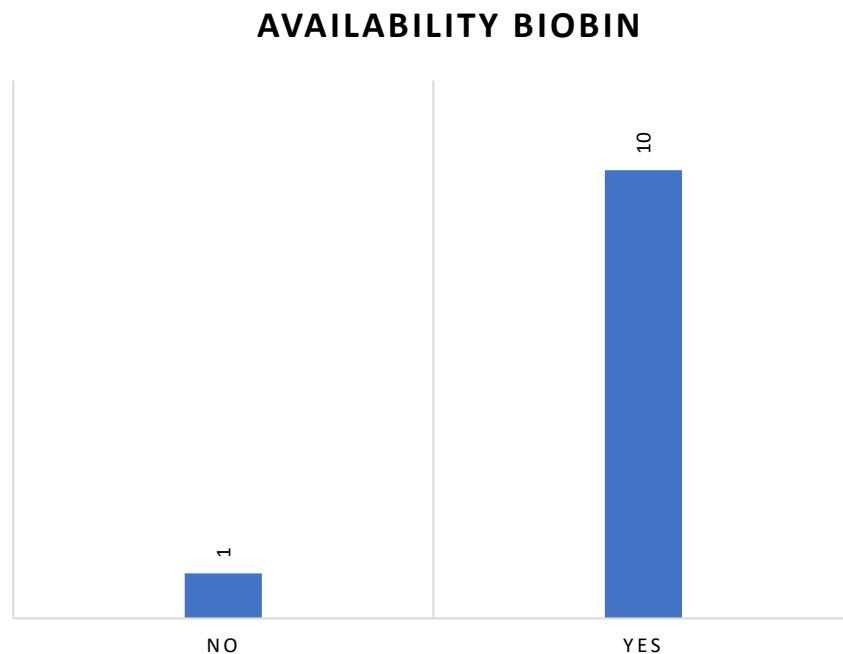


Figure 3.1.6.a Bio digester bin distribution in each panchayath

Inference

10 out 11 panchayat visited had said they started the distribution of bio bin.

b. Comparison between average percentage of households who got Biobin and average number of house holds

Table 3.1.6.b Descriptive statistics of biobin distribution

	Percentage of houses given the biobin	No Houses
Mean	8.8681	8719.727
Std. Deviation	10.19756	2064.488

Inference

There is an average of 8720 houses in a panchayat at Thrissur district and an average 9% of houses received the bio bin.

c. Usage of Bio digester Bin among public

From the response of the public the frequency table of the people who received bio bin is given below.

Table 3.1.6.c Usage of Bio digester Bin among public

Did you get the Biodigester Bin	Frequency	Percent
No	467	83.096
Yes	95	16.904
Total	562	100.000

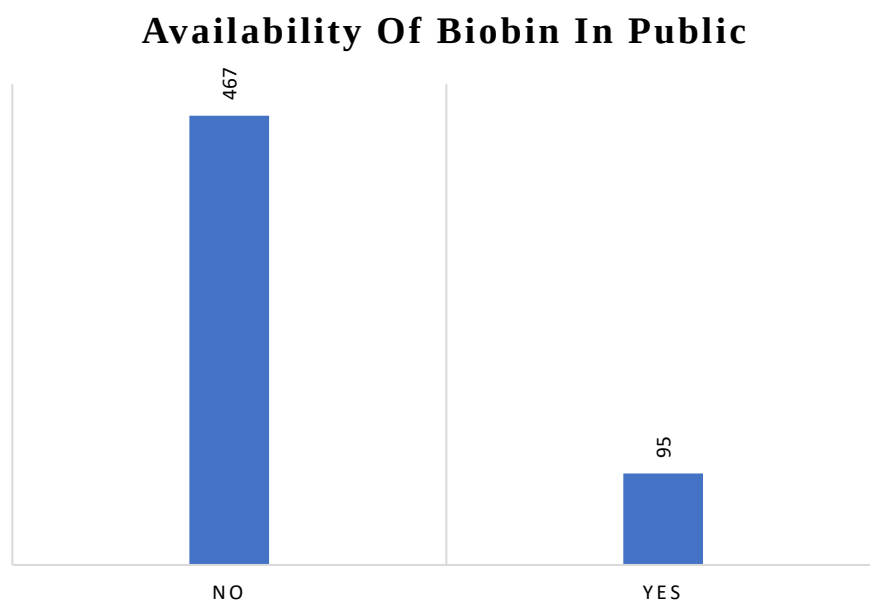


Figure 3.1.6.c Usage of Bio digester Bin among public

Inference

83% of the public have not received the biodigester bin from panchayat and 17% public have received it.

d. Efficient use of bio bin among public

From those who received bio bin, frequency of people who are using it efficiently.

Table 3.1.6.d Efficient use of bio bin among public

Efficient use of biobin	Frequency	Percent
No	17	55.497
Yes	78	44.503
Total	95	100.000

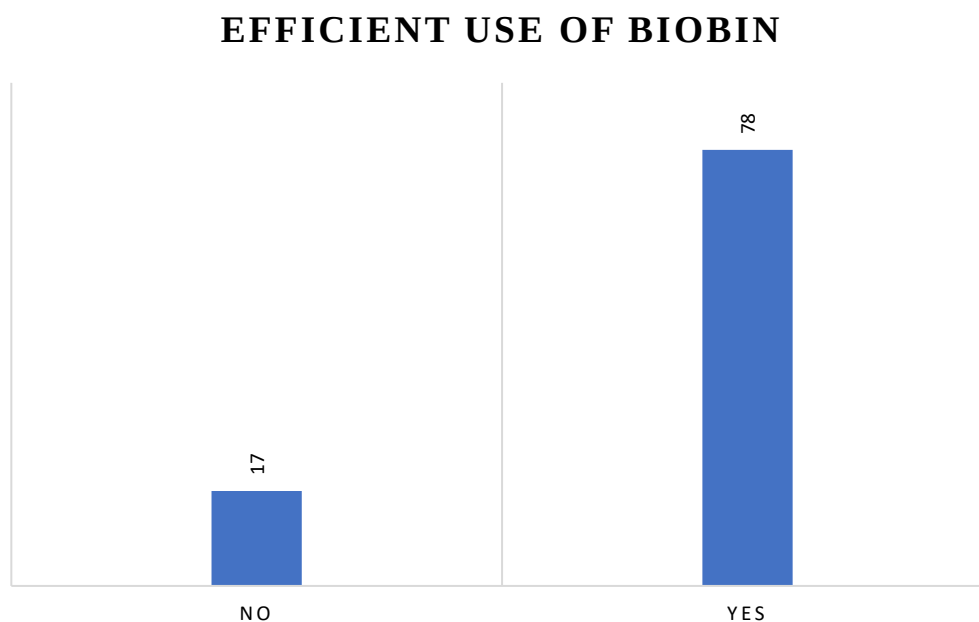


Figure 3.1.6.d Efficient use of bio bin among public

Inference

45% public who received the biodigester bin are using it efficiently.

e. Usage of Bio digester bin among HKS

From the response Haritha karma sena the frequency table of the HKS members who received bio bin is given below.

Table 3.1.6.e Usage of Bio digester bin among HKS

do you use biodigester bin effectively	Frequency	Percent
No	93	54.386
Yes	65	38.012
No Response	13	7.602
Total	171	100.000

EFFICIENT USE BY HARITHA KARMA SENA

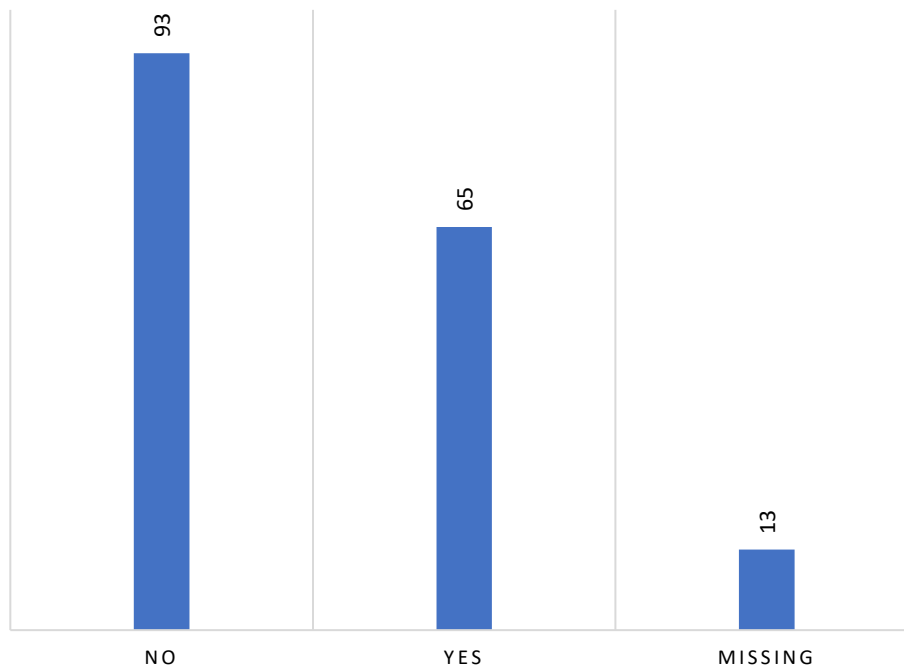


Figure 3.1.6.e Usage of Bio digester bin among HKS

Inference

54% of HKS members received biodigester bin and they are using it efficiently where 38% did not receive it.

3.2 Statistical Tests

3.2.1 TEST 1

TO TEST THE SIGNIFICANT ASSOCIATION BETWEEN HARITHAMITRAM APP USAGE AND THE OPINION THAT THE APP MAKES WORK EASIER

H₀: There is no significant association between the Haritha mitram app usage and the opinion that the app makes work easier for HKS members.

H₁: There is a significant association between the Haritha mitram app usage and the opinion that the app makes work easier for HKS members.

Table 3.2.1.1 Frequency table

	No	Yes
No	17	9
Yes	19	98

Pearson's Chi-squared test with Yates' continuity correction

Table 3.2.1.2 Chi-square table

Test statistic	df	P value
24.73	1	6.599e-07 * * *

Inference

Since the p-value is less than 0.05, we failed to reject the null hypothesis. Hence, there is a significant association between Haritha mitram app usage and the opinion that the app makes work easier for HKS members. The Haritha mithram app improves the quality of work for HKS members.

3.2.2 TEST 2

TO TEST IF THERE IS ANY DIFFERENCE IN THE FREQUENCY OF MEDICAL CHECKUPS TAKING UP THE JOB AS AN HKS MEMBER

H₀: There is no significant difference in the frequency of medical checkups before and after taking up the job as an HKS member.

H₁: There is a significant difference in the frequency of medical checkups before and after taking up the job as an HKS member.

Table 3.2.2.1 Frequency table

	No	Yes
0	63	6
1	42	18
2	13	15
3	6	4
4	0	2
5	1	1

Wilcoxon signed rank test with continuity correction

Table 3.2.2.2 Wilcoxon table

Test statistic	P value	Alternative hypothesis
66	0.003822 * *	two. sided

Inference

Since the p-value is less than 0.05, we reject the null hypothesis. Hence, there is a significant difference in the frequency of medical checkups before and after taking up the job as an HKS member.

3.2.3 TEST 3

TO TEST THE ASSOCIATION BETWEEN WORKING AS A HKS MEMBER AND INCREASE IN THE FREQUENCY OF MEDICAL CHECKUPS

H₀: Working in this field as a HKS member has no significant effect on the occurrence of health problems or the frequency of medical checkups.

H₁: Working in this field as a HKS member has a significant effect on the occurrence of health problems or the frequency of medical checkups.

Table 3.2.3.1 Frequency table

	No	Yes
No	66	5
Yes	59	41

Pearson's Chi-squared test with Yates' continuity correction

Table 3.2.3.2 Chi-square table

Test statistic	df	P value
22.65	1	1.942e-06 * * *

Inference

Since the p-value is less than 0.05, we reject the null hypothesis. i.e., working in this field as a HKS member has a significant effect on the occurrence of health problems or the frequency of medical checkups.

3.2.4 TEST 4

TO TEST PUBLIC BEHAVIOR TOWARDS HKS MEMBERS BASED ON THEIR LEVEL OF WASTE MANAGEMENT AWARENESS.

H₀: There is no significant difference in public behavior towards HKS members based on their level of waste management awareness.

H₁: There is a significant difference in public behavior towards HKS members based on their level of waste management awareness.

Pearson's Chi-squared test: waste management awareness of public and public behavior towards Haritha karma sena

Table 3.2.4.1 Chi-square table

Test statistic	df	P value
3.323	4	0.5054

Inference

Since the p-value is greater than 0.05, we failed to reject the null hypothesis. Hence, there is no significant difference in public behavior towards HKS members based on their level of waste management awareness.

3.2.5 TEST 5

TO TEST THE RELATIONSHIP BETWEEN EDUCATIONAL LEVEL AND HARITHAMITHRAM APP EFFECTIVENESS

H_0 : There is no relationship between a person's educational level and whether the app makes their job simpler.

H_1 : There is a relationship between a person's educational level and if the app makes their job simpler.

Table 3.2.5.1 Frequency table

	No	Yes
Below SSLC	13	38
SSLC	12	42
Plus 2	8	18
Degree	1	6
Others	2	2

Fisher's Exact Test for Count Data

Table 3.2.5.2 Fisher's exact table

P value	Alternative hypothesis
0.6442	two.sided

Inference

Since the p-value is greater than 0.05, we failed to reject the null hypothesis. There is no relationship between a person's educational level and whether the app makes their job simpler. Hence This suggests that the app might be equally useful at different educational levels.

3.2.6 TEST 6

TO TEST SATISFACTION WITH HKS WASTE MANAGEMENT SERVICES AMONG INDIVIDUALS

H_0 : There is no significant association between people who are aware about waste management and their satisfaction with HKS waste management services.

H_1 : There is a significant association between people who are aware about waste management and their satisfaction with HKS waste management services.

Table 3.2.6.1 Frequency table

	Dissatisfied	Neutral	Satisfied	Very Dissatisfied	Very satisfied
No	8	33	97	3	55
Yes	20	51	155	1	139

Pearson's Chi-squared test

Table 3.2.6.2 Chi-square table

Test statistic	df	P value
9.132	4	0.05787

Inference

Since the p-value is greater than 0.05, we failed to reject the null hypothesis. Hence, there is no significant association between people who are aware about waste management and their satisfaction with HKS waste management services.

3.2.7 TEST 7

TO TEST THE RELATIONSHIP BETWEEN AWARENESS OF HARITHA KERALA MISSION AIMS AND SATISFACTION WITH HARITHA KARMA SENA SERVICES

H₀: There is no significant relationship between awareness of the Haritha Kerala Mission's aims and satisfaction with Haritha Karma Sena's services.

H₁: There is no significant relationship between awareness of the Haritha Kerala Mission's aims and satisfaction with Haritha Karma Sena's services.

Table 3.2.7.1 Frequency table

	Dissatisfied	Neutral	Satisfied	Very Dissatisfied	Very satisfied
No	12	41	86	3	63
Yes	16	43	166	1	131

Pearson's Chi-squared test:

Table 3.2.7.2 Chi-square table

Test statistic	df	P value
10.51	4	0.03267 *

Inference

Since the p-value is greater than 0.05, we failed to reject the null hypothesis. Hence, there is no significant relationship between awareness of the Haritha Kerala Mission's aims and satisfaction with Haritha Karma Sena's services.

3.2.8 TEST 8

TO TEST THE ASSOCIATION BETWEEN THE HARITA KARMA SENA MONTHLY SERVICES AND THE DETECTION OF ILLEGAL GARBAGE DUMPING IN THE PANCHAYATH

H₀: There is no significant association between the Haritha Karma Sena monthly services and the detection of illegal garbage dumping in the panchayath.

H₁: There is a significant association between the Haritha Karma Sena monthly services and the detection of illegal garbage dumping in the panchayath.

Table 3.2.8.1 Frequency table

	No	Yes
No	16	7
Yes	411	128

Pearson's Chi-squared test with Yates' continuity correction: table7

Table 3.2.8.2 Chi-square table

Test statistic	df	P value
0.2362	1	0.627

Inference

Since the p-value is greater than 0.05, we failed to reject the null hypothesis. Hence, there is no significant association between the Haritha Karma Sena monthly services and the detection of illegal garbage dumping in the panchayath

Chapter 4

SUMMARY

Among 86 Panchayats in Thrissur district 11 panchayats was randomly selected and the data was collected about Haritha karma Sena. Statistical Analysis was performed on the data. Based on our study, about Haritha Karma Sena, the following are the conclusions. Satisfaction level of the Haritha Karma Sena with the way they are treated by the public was studied. Only 12% of Haritha Karma Sena are very satisfied with the public reaction. 74% of them have experienced reluctance from public while collecting waste. Most of the workers belong to the age category 50-60. Many of the members have been engaged in other jobs before becoming a Haritha Karma Sena member. We studied about the health issues of Haritha Karma Sena after taking up this job. 60% of the workers experienced health issues after engaging in this work, one of the major issue was skin problems like itching. Most of them work 25-30 days in a month and 8 hours in a day. Then we studied the satisfaction of the public with the activities of Haritha Karma Sena, only 45% of the people are satisfied with their activities. By analysing the awareness of people about the goals of Haritha Karma Sena, we understood that 64% of them have the knowledge about the goals of Haritha Karma Sena. Haritha Karma Sena also tried to educate people regarding their services. Most of the workers of Haritha Karma Sena used the Haritha Mithram app effectively. The app makes their job easy. But they also face some technical issues regarding the range while using the app. The panchayat provides biodigester bin to people for the disposal of biodigestible wastes. Based on our study we understood that most of the people didn't get the biodigester bin. Only 16% got the bin. Also in a panchayat an average of 9% houses received the bio bin where there is average 8720 houses in each panchayat and Madakkathara panchayat did not started the distribution of bio bin.

Conclusion

The project highlights several critical aspects of the Haritha Karma Sena's operations and challenges. While there is a significant proportion of satisfaction among the workers regarding public treatment, a notable percentage report dissatisfaction and reluctance from the public. This indicates a need for better public awareness and cooperation.

The demographic data show that many members are older, which may contribute to the physical challenges and health issues reported. The increase in hospital visits and health problems among workers underscores the need for improved safety and health measures.

Public knowledge about the Haritha Kerala Mission and the efficient use of biodigester bins also requires enhancement to ensure the mission's success. Not all panchayats started the distribution of bio digester bins so that the people could not dispose the bio degradable waste properly.

Addressing these issues through targeted interventions and increased community engagement can improve both the effectiveness of the Harita Karma Sena and the overall waste management efforts in the region.

Suggestions from public

- The Government's schedule should be followed correctly while collecting the non-bio degradable wastes and schedule or calendar should be made available to the public also.
- Dispose of the waste correctly and provide waste collection bins to each house and establishments.
- Collecting wastes from the public places with collaborating other social clubs will be more effective to reduce wastes.
- If there is no waste, there is no need to collect money and evaluate the price based on the quantity of the waste.
- Make sure that the Haritha karma sena members are wearing mask, gloves like safety equipment as a precaution to illness.

APPENDIX

Questionnaire

I ഉദ്യോഗസ്ഥരോട് ചോദിക്കേണ്ട ചോദ്യങ്ങൾ

1. പഞ്ചായത്തിന്റെ പേര്
2. പഞ്ചായത്തിലെ ആകെ വാർഡുകളുടെ എണ്ണം.
3. ഓരോ വാർഡുകളിലെയും ആകെ വീടുകളുടെ എണ്ണം
4. ഓരോ വാർഡുകളിലെയും ആകെ സ്ഥാപനങ്ങളുടെ എണ്ണം
5. ഓരോ വാർഡുകളിലും പ്രവർത്തിക്കുന്ന ഹരിത കർമ്മ സേന അംഗങ്ങളുടെ എണ്ണവും പേരു വിവരങ്ങളും (Phn no)
6. പ്ലാസ്റ്റിക് പൊതുസ്ഥലങ്ങളിൽ നിക്ഷേപിക്കുകയും കത്തിക്കുകയും ചെയ്യുന്നതിനെ സംബന്ധിച്ചു പരാതികൾ ലഭിക്കാറുണ്ടോ?
*ഉണ്ട്
*ഇല്ല
7. ഉണ്ടെങ്കിൽ എത്ര?
8. വാർഡിലെ എല്ലാ വീടുകളും സ്ഥാപനങ്ങളും ഹരിത കർമ്മ സേനയോടു സഹകരിക്കാറുണ്ടോ?
*ഉണ്ട്
*ഇല്ല
9. ഇല്ലെങ്കിൽ പ്ലാസ്റ്റിക് അല്ലാത്ത legacy waste ഉന്മൂലനം ചെയ്യുന്ന മാർഗം അന്വേഷിക്കാറുണ്ടോ?
*ഉണ്ട്
*ഇല്ല
10. ഇതുവരെ ഓരോ മാസങ്ങളിലായി എത്ര വീടുകളിൽ ഹരിത കർമ്മ സേനയുടെ സേവനം എത്തിയിട്ടുണ്ട്.
11. ഇതുവരെ എല്ലാ വാർഡുകളിലെയും വീടുകളിൽ നിന്നും ശേഖരിച്ചിട്ടുള്ള മാലിന്യങ്ങളുടെ അളവ്
12. എല്ലാ വീടുകളിൽ നിന്നും സേവനത്തിന്റെ നിശ്ചിത തുക കിട്ടാറുണ്ടോ
*ഉണ്ട്
*ഇല്ല

13. ഓരോ മാസങ്ങളിലും എത്ര വീടുകളിൽ നിന്ന് സേവനത്തിന്റെ നിശ്ചിത തുക ലഭ്യമായിട്ടുണ്ട്?
14. ഇതുവരെ ഓരോ മാസങ്ങളിലായി എത്ര സ്ഥാപനങ്ങളിൽ ഹരിത കർമ്മ സേനയുടെ സേവനം എത്തിയിട്ടുണ്ട്.
15. ഇതുവരെ എല്ലാ വാർഡുകളിലെയും സ്ഥാപനങ്ങളിൽ നിന്നും ശേഖരിച്ചിട്ടുള്ള മാലിന്യങ്ങളുടെ അളവ്
16. എല്ലാ സ്ഥാപനങ്ങളിൽ നിന്നും സേവനത്തിന്റെ നിശ്ചിത തുക കിട്ടാറുണ്ടോ
*ഉണ്ട്
*ഇല്ല
17. ഓരോ മാസങ്ങളിലും എത്ര സ്ഥാപനങ്ങളിൽ നിന്ന് സേവനത്തിന്റെ നിശ്ചിത തുക ലഭ്യമായിട്ടുണ്ട്?
18. അധിക തുക കിട്ടാറുണ്ടോ?
*ഉണ്ട്
*ഇല്ല
19. ഗവൺമെന്റ് നൽകിയിരിക്കുന്ന കലണ്ടർ പ്രകാരമാണോ പ്ലാസ്റ്റിക് അല്ലാത്ത മാലിന്യങ്ങൾ ശേഖരിക്കാറുള്ളത്?
*അതെ
*അല്ല
20. പഞ്ചായത്തിൽ എത്ര MCF കൾ ഉണ്ട്?
21. പഞ്ചായത്തിൽ എത്ര mini MCF കൾ ഉണ്ട്?
22. പഞ്ചായത്തിൽ RRF പ്രവർത്തിക്കുന്നുണ്ടോ?
23. ഓരോ മാസങ്ങളിലും ഉപയോക്താവ് രേഖപ്പെടുത്തിയിട്ടുള്ള പരാതികളുടെ എണ്ണം
24. അതിൽ പരിഹരിക്കപ്പെട്ട പരാതികളുടെ എണ്ണം

II ഹരിത കർമ്മ സേനയോട് ചോദിക്കേണ്ട ചോദ്യങ്ങൾ

1. വയസ്സ്
 1. 40 വരെ
 2. 40-50
 3. 50-60
 4. 60-70

5. 70 ന് മുകളിൽ

2.നിങ്ങൾ മുമ്പ് മറ്റെന്തെങ്കിലും ജോലി ചെയ്തിട്ടുണ്ടോ?

- a.ഉണ്ട്
- b.ഇല്ല

3.താഴെപ്പറയുന്നവയിൽ എന്തെല്ലാമാണ് പഞ്ചായത്തിൽ നിന്നോ ബന്ധപ്പെട്ട അധികാരികളിൽ നിന്നോ നിങ്ങൾക്ക് ലഭിക്കുന്നത്?

- 1. ഇൻഷുറൻസ്
- 2. വാഹനം
- 3. മാസ്കും മറ്റ് സുരക്ഷാ ഉപകരണങ്ങളും
- 4. യൂണിഫോം
- 5. ഐ ഡി കാർഡ്

4.ഒരു മാസത്തിൽ എത്ര തവണ നിങ്ങൾ വൈദ്യപരിശോധനയ്ക്കായി ഡോക്ടറെ സന്ദർശിക്കാറുണ്ട്?

5.ഈ ജോലി ഏറ്റെടുത്ത ശേഷം മെഡിക്കൽ ചെക്കപ്പുകളുടെ ആവൃത്തിയിൽ എന്തെങ്കിലും വർദ്ധനവ് നിങ്ങൾ ശ്രദ്ധിച്ചിട്ടുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

6.ഈ ജോലി ഏറ്റെടുത്തതിനു ശേഷം നിങ്ങൾക്ക് സ്ഥിരമായി എന്തെങ്കിലും ശാരീരിക ബുദ്ധിമുട്ട് നേരിടേണ്ടി വരുന്നുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

7.ഉണ്ടെങ്കിൽ അത് എന്താണെന്ന് വ്യക്തമാക്കാമോ?

- A. ചുമ
- B. ത്വക്ക് രോഗം
- C. നടുവേദന
- D. മറ്റുള്ളവ

8.വീടുവീടാന്തരം കയറി പ്ലാസ്റ്റിക്കും ഫണ്ടും ശേഖരിക്കുമ്പോൾ പൊതുജനങ്ങളിൽ നിന്ന് എന്തെങ്കിലും എതിർപ്പ് നേരിടേണ്ടിവരുന്നുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

9.മാസത്തിൽ ശരാശരി എത്ര ദിവസമാണ് നിങ്ങൾക്ക് വീടുകളിൽ പോയി പ്ലാസ്റ്റിക്കും മറ്റും ശേഖരിക്കാനും അത് മെറ്റീരിയൽ ശേഖരണ കേന്ദ്രത്തിൽ(MCF)ൽ എത്തിച്ച് തരം തിരിക്കാനും ആയി വേണ്ടിവരുന്നത് ?

- 1. 20 തിൽ താഴെ
- 2. 20-25
- 3. 25-30

10. പ്ലാസ്റ്റിക് മറ്റും ശേഖരിക്കുമ്പോൾ പൊതുജനങ്ങളിൽ നിന്നും എന്തെങ്കിലും മോശം അനുഭവം നേരിട്ടിട്ടുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

11. നേരിടേണ്ടി വന്നിട്ടുണ്ടെങ്കിൽ അത് ആരിൽ നിന്നാണെന്ന് വ്യക്തമാക്കാമോ?

- a. വീടുകളിൽ നിന്ന്
- b. മറ്റു സ്ഥാപനങ്ങളിൽ നിന്ന്

12. നിങ്ങൾ സാധാരണയായി ഒരു ദിവസം എത്ര മണിക്കൂർ ജോലി ചെയ്യുന്നു?

13. നിങ്ങൾ നൽകുന്ന സേവനത്തെക്കുറിച്ച് പൊതുജനങ്ങളെ സജീവമായി ബോധവൽക്കരിക്കുന്നുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

14. ഹരിതമിത്രം ആപ്പ് കാര്യക്ഷമമായി ഉപയോഗിക്കാൻ നിങ്ങൾക്ക് സാധിക്കാറുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

15. ഹരിതമിത്രം ആപ്പ് ഉപയോഗിക്കുമ്പോൾ എന്തെങ്കിലും സാങ്കേതിക തടസ്സം നേരിടേണ്ടി വരാറുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

16. ഹരിത മിത്രം ആപ്പ് നിങ്ങളുടെ ജോലി എളുപ്പമാക്കുന്നുണ്ടോ?

- a. ഉണ്ട്
- b. ഇല്ല

17. പൊതുജനങ്ങൾ നിങ്ങളോട് പെരുമാറുന്ന രീതിയിൽ നിങ്ങൾ സംതൃപ്തരാണോ?

- a) വളരെ സംതൃപ്തി
- b) സംതൃപ്തി
- c) ന്യൂട്രൽ
- d) അസംതൃപ്തി
- e) വളരെ അസംതൃപ്തി

18. വിദ്യാഭ്യാസ യോഗ്യത

- a. SSLC ക്ക് താഴെ
- b. SSLC
- c. പ്ലസ് 2
- d. ഡിഗ്രി
- e. മറ്റുള്ളവ

III പൊതുജനങ്ങളോട് ചോദിക്കേണ്ട ചോദ്യങ്ങൾ

1.പ്രതിമാസം, ഹരിത കർമ്മ സേനയുടെ സേവനം ലഭിക്കുന്നുണ്ടോ?

*ഉണ്ട്

*ഇല്ല

2.ഹരിത കർമ്മ സേനയുടെ സേവനത്തിൽ നിങ്ങൾ സംതൃപ്തനാണോ?

*ഉയർന്ന സംതൃപ്തി

* സംതൃപ്തി

*നിഷ്പക്ഷത

*അസംതൃപ്തി

*കടുത്ത അസംതൃപ്തി

3.ഹരിത കർമ്മ സേനയിൽ നിന്ന് നിങ്ങൾക്ക് എപ്പോഴെങ്കിലും മോശം അനുഭവം ഉണ്ടായിട്ടുണ്ടോ?

*ഉണ്ട്

*ഇല്ല

4.ഹരിത കർമ്മ സേന ആരംഭിച്ചതിനു ശേഷവും നിങ്ങളുടെ പ്രദേശത്ത് നിയമവിരുദ്ധമായ മാലിന്യ കുമ്പാരം കാണുന്നുണ്ടോ?

*ഉണ്ട്

*ഇല്ല

5.മാലിന്യ നിർമാർജ്ജനവുമായി ബന്ധപ്പെട്ട് കുറഞ്ഞത് 3 പിഴകൾ നിങ്ങൾക്ക് അറിയാമോ?

*അറിയാം

* ഇല്ല

6.ഹരിത കേരള മിഷന്റെ ലക്ഷ്യങ്ങളെ കുറിച്ച് നിങ്ങൾ ബോധവൽക്കരിക്കാമോ?

*അതെ

*ഇല്ല

7.ഹരിത കർമ്മ സേന എപ്പോൾ എത്തുമെന്ന് ആരെങ്കിലും നിങ്ങളെ അറിയിക്കാറുണ്ടോ ?

*ഉണ്ട്

*ഇല്ല

8.എങ്കിൽ ആരിൽ നിന്നാണ് നിങ്ങൾ അറിയുന്നത്?

*ഹരിത കർമ്മ സേന

*കുടുംബശ്രീ

*പഞ്ചായത്ത് അംഗം

*ആശാ വർക്കർ

*മറ്റുള്ളവർ

9.ഹരിത കർമ്മ സേനയോട് ആരെങ്കിലും മോശമായി പെരുമാറുന്നത് കണ്ടിട്ടുണ്ടോ?

*ഉണ്ട്

*ഇല്ല

10.ഹരിത കർമ്മ സേന ചെയ്യുന്ന പ്രവർത്തനത്തിന് എന്തെങ്കിലും നിർദ്ദേശങ്ങൾ പറയാനുണ്ടോ? എങ്കിൽ എന്ത്?

Chapter 5

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